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Course Name:5G 6G Communication Systems

Course Code:ECA5601

Ex:7 Analog Beamforming Analysis Using MATLAB

Aim:

To analyze the performance of analog beamforming by studying beamforming gain, beam width, and main lobe direction using MATLAB.

Objective:

1. To analyze the beamforming gain achieved by varying the number of antenna elements in an analog beamforming system using MATLAB.
2. To compare the beam width of an antenna array for different beamforming configurations using MATLAB.
3. To analyze the main lobe direction by steering the antenna array beam to different angles using MATLAB.

Software Required:

- MATLAB

Theory

Analog beamforming is a beam steering technique in which a single RF chain drives multiple antenna elements through adjustable phase shifters. By controlling the phase of the transmitted or received signals, the antenna array forms a directional beam toward the intended user, thereby improving signal strength, coverage, and spectral efficiency. Analog beamforming is widely employed in millimeter-wave (mmWave) communication systems due to its reduced hardware complexity and lower power consumption.

Objective 1: Matlab Code and Output

```
clc;
```

```
clear;
```

```
close all;
```

```

% Number of antenna elements
antenna_sizes = [2 4 8 16 32];

% Beamforming Gain (dB)
% Gain = 10*log10(N)
beamforming_gain = 10*log10(antenna_sizes);

% Round values (optional)
beamforming_gain = round(beamforming_gain);

% Create Bar Graph
figure;
bar(antenna_sizes, beamforming_gain, 0.6);

% Title and Labels
title('Beamforming Gain for Different Antenna Array Sizes');
xlabel('Number of Antenna Elements');
ylabel('Beamforming Gain (dB)');

% Axis Limits
ylim([0 16]);
xlim([1 34]);

% Grid
grid on;

% Display Gain Values on Top of Bars
for i = 1:length(antenna_sizes)
    text(antenna_sizes(i), beamforming_gain(i)+0.3, ...

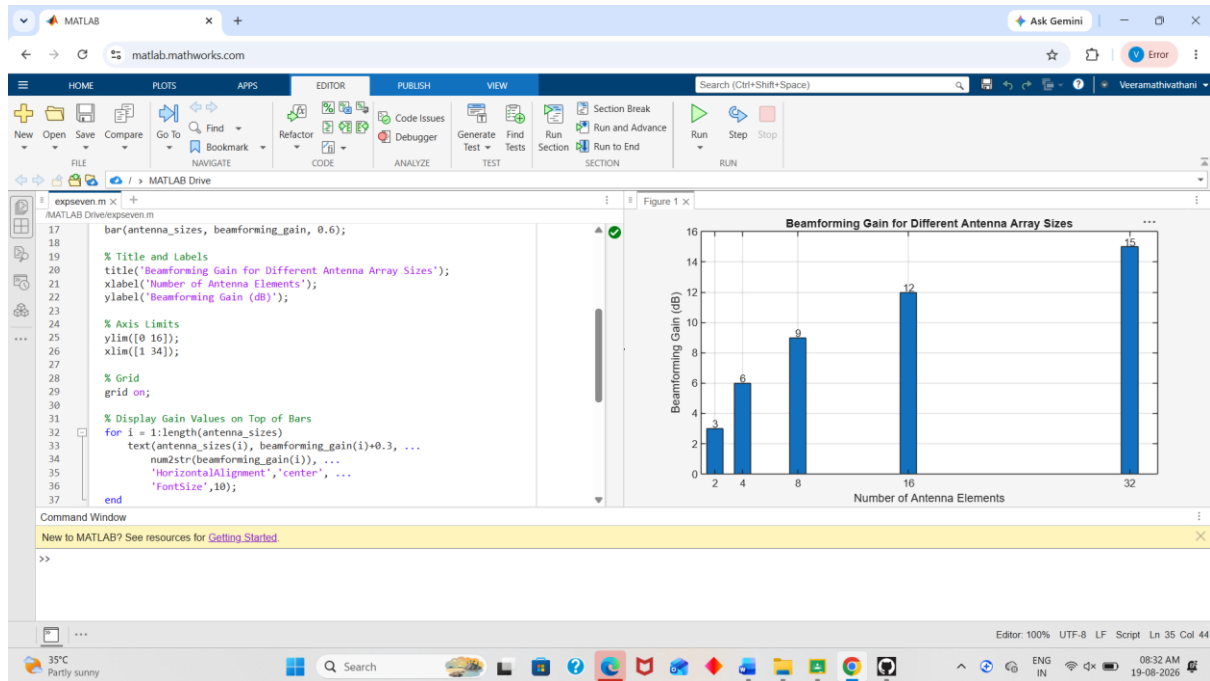
```

```
num2str(beamforming_gain(i)), ...
```

```
'HorizontalAlignment','center', ...
```

```
'FontSize',10);
```

```
End
```



Objective 2: Matlab Code and Output

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)
```

```
figure
```

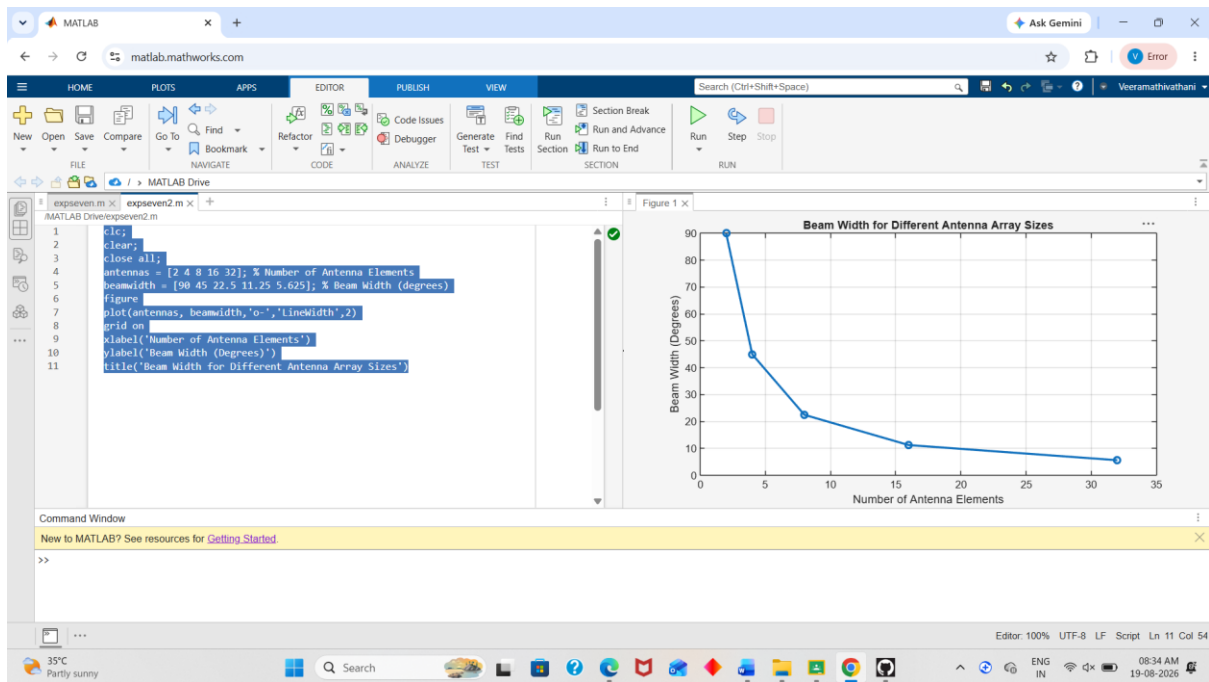
```
plot(antennas, beamwidth,'o-','LineWidth',2)
```

```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beam Width (Degrees)')
```

```
title('Beam Width for Different Antenna Array Sizes')
```



Objective 3: Matlab Code and Output

clc;

clear;

close all;

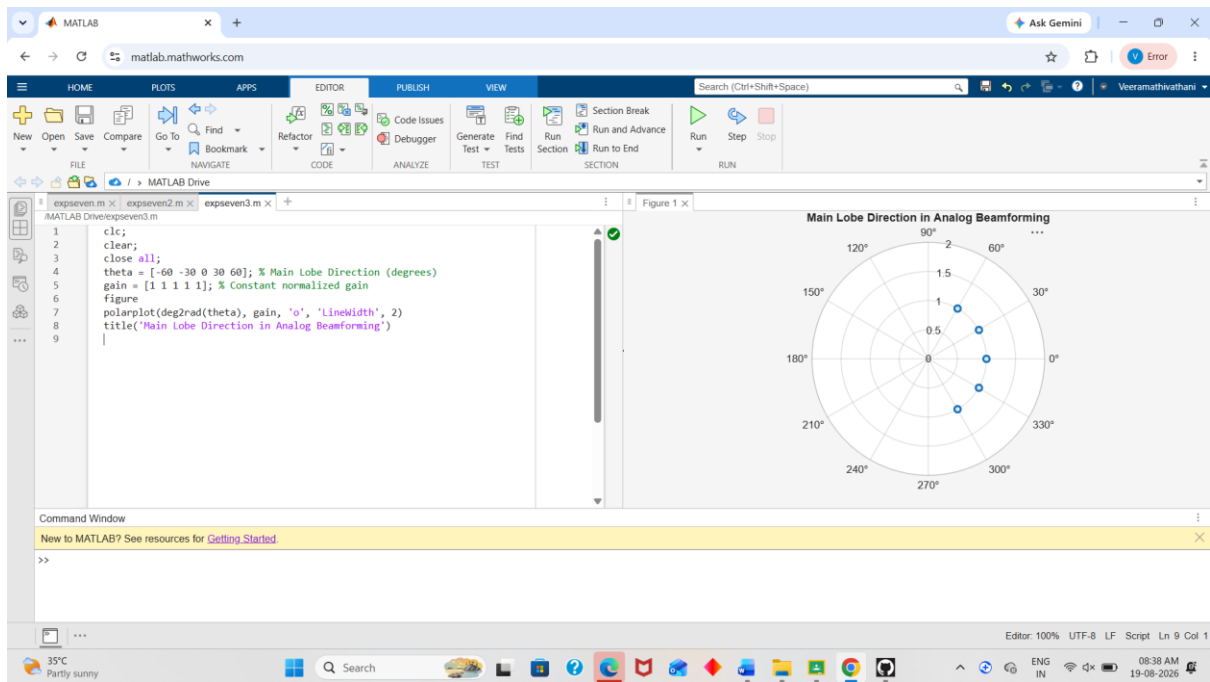
theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)

gain = [1 1 1 1 1]; % Constant normalized gain

figure

polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)

title('Main Lobe Direction in Analog Beamforming')



Result:

The MATLAB analysis successfully demonstrated the characteristics of analog beamforming. The results showed that increasing the number of antenna elements improves beamforming gain and reduces beam width, while phase adjustment enables the main lobe to be steered toward different directions.